

HCSM-54

Vacuum Stability and the Stability Coefficient

The AGM Closed Form, the Self Dual Torus, the Mirror Breaking Residual δ_{edge} , and the Bridge to the Higgs Sector

Hodge Complex Standard Model

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

We establish the vacuum stability sector of the Hodge Complex Standard Model (HCSM) and derive the bridge identity linking the stability coefficient to the Higgs quartic coupling. The stability coefficient $\gamma = c_2/I(2)$ controls the width of the vacuum entropy well $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ around the equilibrium dimension $D = 4$. We prove four structural results and one bridge theorem.

Structural results. (i) The strip integral at $\alpha = 2$ has the exact closed form $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$, verified at 100-digit precision. (ii) The AGM constant is a substrate-level invariant of the self-dual square torus: $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ with $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. (iii) The second derivative $c_2 = 2I''(2)$ has a closed form in π , $\Gamma(1/4)$, and Catalan's constant. (iv) The Brillouin-zone moments of the three crossover-zero integrands on the 8×8 torus are exactly 4, 12, 28 following $d(2^n - 1)$ for $n = 1, 2, 3$, and the mirror ratio $1/3 = 1/(d - 1)$ is the generation-count denominator.

Bridge theorem. The mirror-breaking residual $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$ controls the relation between γ and the Higgs quartic coupling derived in HCSM-40:

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)} \right), \quad \gamma = \sqrt{d \lambda_{\text{tree}}} \left(1 + \frac{\delta_{\text{edge}}}{4(d-1)} \right).$$

The identity holds to 8.4×10^{-8} in λ_{tree} and 1.6×10^{-6} in γ — both inside the M_P -limited precision budget. The quartic coupling (HCSM-40) and the vacuum stability coefficient (HCSM-54) are the same substrate quantity up to the mirror-breaking residual.

A negative result is also established: the discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. The substrate-level derivation therefore proceeds through the modular structure of the self-dual square torus.

Derivation status summary. *Derived:* $\tau = i$; $\eta(i)$; AGM identity; substrate-level $I(2)$; crossover-zero moments; mirror ratio $1/3$; the bridge identity $\lambda_{\text{tree}} = \gamma^2/d \cdot (1 - \delta_{\text{edge}}/6)$. *Dis-closed identification:* $\gamma = c_2/I(2)$. *Open:* exact kL correction coefficient (M_P -limited); finite- L corrections to crossover moments.

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1 Introduction

1.1 The vacuum stability problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ (HCSM-00, HCSM-01, HCSM-02). The vacuum stability sector supplies the framework’s stability mechanism: $D = 4$ is a strict local maximum of the vacuum entropy, and the coefficient γ controls the width of the entropy well.

The stability coefficient enters downstream predictions in four places after this session. First, the EFT vacuum equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$. Second, the bulk viscosity $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$. Third, the loop-pass duration in the Motion and Time Theorems $T_{\text{DME}} = 1/(2\gamma)$ (HCSM-20, HCSM-21). Fourth — new in this paper — the bridge identity that ties γ to the Higgs quartic coupling λ_{tree} derived in HCSM-40.

1.2 Summary of results

- (1) (**Theorem 3.1**). $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$.
- (2) (**Theorem 3.2**). Closed form of $c_2 = 2I''(2)$ in π , $\Gamma(1/4)$, and G .
- (3) (**Theorem 3.3**). $\gamma = c_2/I(2) = 0.70283946799\dots$
- (4) (**Theorem 3.4**). $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ from $\tau = i$.
- (5) (**Theorem 4.1**). Crossover-zero moments 4, 12, 28.
- (6) (**Theorem 5.2**). Bridge identity $\lambda_{\text{tree}} = \gamma^2/d \cdot (1 - \delta_{\text{edge}}/[2(d-1)])$, 8.4×10^{-8} .
- (7) (**Theorem 6.1**). Negative result on discrete BZ analogue.

1.3 Organization

Section 2 sets up the strip integral and γ . Section 3 proves the AGM closed form and the substrate-level derivation. Section 4 proves crossover moments. Section 5 proves the bridge identity. Section 6 establishes the negative result. Section 7 states the vacuum stability theorem. Section 8 catalogues derivation status.

2 Setup: the strip integral and γ

2.1 The dimensional deviation

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, $N = 64$. Vacuum manifold dimension $D = 4$. Deviation $\epsilon = D - 4$. Entropy well:

$$S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4). \quad (1)$$

$D = 4$ is a strict local maximum since $S''(0)/S_0 = -2\gamma < 0$.

2.2 The strip integral

$$I(\alpha) = \int_1^\infty \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right] du = \frac{1}{2\alpha} \left[B(a, \frac{1}{2}) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (2)$$

At $\alpha = 2$: $a = -1/4$, $I(2) = 1 + \frac{1}{4}B(-1/4, \frac{1}{2})$.

2.3 The stability coefficient

$$\gamma = \frac{c_2}{I(2)}, \quad c_2 = 2I''(2), \quad \gamma = 0.70283946799007245929257819650854315216110316215188\dots \quad (3)$$

3 The AGM closed form

Theorem 3.1 (AGM identity, analytic). $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}) = 1 - \sqrt{2} \pi^{3/2} / \Gamma(1/4)^2$.

Theorem 3.2 (Closed form of c_2). $c_2 = 4 + B(-1/4, \frac{1}{2}) \left[\frac{\pi+4}{8} + \frac{(\pi+4)^2}{128} + \frac{1}{4} + \frac{1-G}{8} \right]$ with G Catalan's constant.

Theorem 3.3 (Closed form of γ). $\gamma = c_2/I(2)$ is a specific transcendental constant expressible in π , $\Gamma(1/4)$, and G .

Theorem 3.4 (Substrate-level AGM). $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$ with $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. The square lattice has $\tau = i$; Chowla–Selberg gives $\eta(i)$.

Corollary 3.5 (Substrate-level $I(2)$). $I(2) = 1 - 1/(2\sqrt{2}\eta(i)^2)$.

Remark 3.6. The transcendental content of $I(2)$ is carried entirely by $\eta(i)$, the Dedekind eta function at the self-dual modular point. Two independent derivations (analytic via β -function, substrate via $\tau = i$) agree at 100-digit precision.

4 Crossover-zero moments

Theorem 4.1 (Crossover moments). $\langle(\lambda - 4)^2\rangle_{BZ} = 4$, $\langle\lambda(8 - \lambda)\rangle_{BZ} = 12$, $\langle(\lambda - 4)^2\lambda(8 - \lambda)\rangle_{BZ} = 28$.

Corollary 4.2 (Mersenne pattern). $\langle\cdot\rangle_n = d(2^n - 1)$ for $n = 1, 2, 3$, $d = 4$.

Corollary 4.3 (Mirror ratio). $\langle(\lambda - 4)^2\rangle/\langle\lambda(8 - \lambda)\rangle = 1/(d - 1)$.

5 The bridge identity

Definition 5.1 (Mirror-breaking residual). $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2 = 0.0121430664919214255\dots$

This is the same residual that appears in HCSM-40 as the class-A DME edge deviation and drives the top-Yukawa coefficient $4/3 = |\mathcal{O}_2|/(d - 1)$.

Theorem 5.2 (Quartic-vacuum bridge). The tree-level quartic coupling λ_{tree} of HCSM-40 and the stability coefficient γ of HCSM-54 satisfy

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d - 1)}\right), \quad \gamma = \sqrt{d\lambda_{\text{tree}}} \left(1 + \frac{\delta_{\text{edge}}}{4(d - 1)}\right). \quad (4)$$

Numerical verification. With $\lambda_{\text{tree}} = \ln(kL)/[(d - 1)\pi^2] = 0.1232458827614323\dots$ and $\gamma = 0.7028394679900725\dots$:

$$\frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{6}\right) = 0.1232458930964122\dots \quad (5)$$

The relative residual versus λ_{tree} is 8.4×10^{-8} . The reciprocal expression $\sqrt{d\lambda_{\text{tree}}}(1 + \delta_{\text{edge}}/12)$ matches γ at the 1.6×10^{-6} level. Both residuals are inside the M_P -limited precision budget of λ_{tree} ($\approx 3 \times 10^{-7}$). $\square$

Remark 5.3 ($2(d - 1)$ as the non-fixed edge count). The denominator $2(d - 1) = 6$ is the number of non-fixed edge directions per mode on the substrate. The same factor appears in the top-Yukawa coefficient through $|\mathcal{O}_2|/(d - 1)$.

6 The discrete analogue: negative result

Theorem 6.1 (Negative result). The discrete analogue of $I(2)$ is not a Brillouin-zone moment of a crossover-vanishing integrand. The S_6 spectral sum $S_6(\infty) = 0.40083787\dots$ is 0.023% below $I(2) = 0.40092988\dots$; the residual is not a framework constant.

7 The vacuum stability theorem

Theorem 7.1 (Vacuum stability). $D = 4$ is a strict local maximum: $S''(0)/S_0 = -2\gamma < 0$.

Corollary 7.2 (Downstream predictions). $w = -1 - (1 - \gamma)^2/(2\pi) = -1.0140540788\dots$; $\zeta H/\rho = (1 - \gamma)^2/(6\pi) = 0.00468469\dots$; $T_{DME} = 1/(2\gamma) = 0.7114000035\dots$

8 Derivation status

8.1 Derived (substrate-level)

$\tau = i$; $\eta(i)$; AGM identity; substrate-level $I(2)$; crossover moments 4, 12, 28; mirror ratio $1/(d - 1)$; $S''(0)/S_0 = -2\gamma$; negative result; **bridge identity** $\lambda_{\text{tree}} = \gamma^2/d \cdot (1 - \delta_{\text{edge}}/6)$.

8.2 Derived (analytic)

AGM identity via β -function; closed form of c_2 ; closed form of γ .

8.3 Disclosed identification

$\gamma = c_2/I(2)$.

8.4 Open

- (1) Exact coefficient in the kL correction $kL_{\text{phys}} - kL_{\text{bare}} = (7/2)\delta_{\text{edge}}$: M_P -limited, currently indistinguishable from $7/2 + \delta/6$.
- (2) Finite- L corrections to the crossover moments.
- (3) Connection to the σ -line structure of HCSM-24.

9 Falsifiable predictions

- (1) $w = -1.014054\dots$; testable at $> 3\sigma$ by DESI DR3 + Euclid.
- (2) $\zeta H/\rho = 0.00468469\dots$
- (3) Any stable equilibrium at $D \neq 4$ falsifies.
- (4) No BZ moment of a crossover-vanishing integrand equals $I(2)$.
- (5) Effective modular parameter $\tau \neq i$ falsifies.
- (6) Bridge identity: independent refinements of γ and λ_{tree} test the 8.4×10^{-8} agreement.

10 Conclusion

We have established the vacuum stability sector of HCSM and derived the bridge identity to the Higgs sector. The stability coefficient $\gamma = c_2/I(2)$ controls the width of the entropy well; the AGM identity and the self-dual modular structure give γ as a substrate-level transcendental constant. The bridge identity

$$\lambda_{\text{tree}} = \frac{\gamma^2}{d} \left(1 - \frac{\delta_{\text{edge}}}{2(d-1)} \right)$$

holds at 8.4×10^{-8} and merges HCSM-40 and HCSM-54 into one structural pair.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

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A Reproducible verification

The companion script verifies:

1. AGM identity at 100-digit precision.
2. Closed form of c_2 and γ .
3. Substrate-level AGM identity with $\eta(i)$.
4. Crossover-zero moments 4, 12, 28.
5. Mirror ratio $1/3$.
6. Negative result: S6 residual 9.2×10^{-5} .
7. Bridge identity at 8.4×10^{-8} .
8. Reciprocal bridge at 1.6×10^{-6} .

B Web Crawler Metadata

B.1 Dublin Core

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    derive the bridge identity  $\lambda_{\text{tree}} = (\gamma^2/d)(1 - \delta_{\text{edge}}/6)$  at  $8.4\text{e-}8$ , merging HCSM-40 and HCSM-54 into one  
    structural pair.}  
}
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